

MD. SHOUROV

Software Engineering Undergraduate · Institute of Information Technology (IIT), University of Dhaka

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Codeforces: Shourov735 (1,500+ Solved)

EDUCATION

Institute of Information Technology (IIT), University of Dhaka — B.S. in Software Engineering

2024 – Present

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Design Patterns, Database Management Systems, Software Requirements Analysis, Computer Architecture, Discrete Mathematics.
- **Admission Record:** Secured top national merit rank in the University of Dhaka Engineering Admission Test (Rank: 735).

EXPERIENCE & LEADERSHIP

IT Secretary & Lead Platform Architect — Nabodigonto Social Welfare Organization

Dec 2024 – Present

- Appointed to executive board as **IT Secretary** after independently architecting and deploying the organization's core digital platform.
- Designed **Blind Audit Protocol & Double-Signed Verification**, cryptographically concealing treasury balances until dual-officer sign-off.
- Built dynamic **3D-flip digital member ID cards** with public QR verification and automated print-ready PVC batch layouts.
- Reduced monthly statement turnaround time by **80%** with automated dues tracking, instant receipt issuance, and SHA-256 verified PDF audit reports.

FEATURED TECHNICAL PROJECTS

Nabodigonto Platform | *Next.js 16, Cloudflare Workers, Neon PostgreSQL, Prisma, Clerk v7, Tailwind CSS*

[Live Demo ↗](#) · [GitHub ↗](#)

- Production community and treasury platform deployed serverless on Cloudflare Workers via OpenNext with global edge caching.
- Integrated multi-tenant dues tracking, automated financial statements, and role-based officer permissions with Clerk auth.

Insidejibon EdTech Platform | *Next.js 16, Cloudflare Workers & R2, Drizzle ORM, Neon PostgreSQL, Clerk Auth*

[Live Demo ↗](#) · [GitHub ↗](#)

- Scalable bilingual (English/Bengali) EdTech web platform supporting course catalogs, lesson delivery, student enrollments, and online exams.
- Integrated Cloudflare R2 object storage for distributed media delivery with sub-100ms global edge latency.

Quiz Management & Examination System | *Java 17, JavaFX 21, SQLite, Design Patterns, JUnit 5, Maven*

[GitHub ↗](#)

- Engineered desktop examination suite for IIT DU Design Patterns Lab (Mini-SPL) implementing **GoF patterns**: *Strategy* (scoring engines), *State* (quiz lifecycle), *Factory* (question authoring: MCQ, True/False, Fill-in, Multi-answer), and *Observer* (countdown & event pub/sub).
- Implemented relational persistence using SQLite via JDBC, automated schema migration, and comprehensive test suite with JUnit 5.

Quran Reading Tracker | *React Native, Expo, SQLite / AsyncStorage, TypeScript, Mobile Android*

[GitHub ↗](#)

- Architected offline-first Android application with zero cloud telemetry, ensuring complete privacy and 100% local persistence.
- Engineered dual-pipeline daily habit tracking for Arabic recitation and Bengali translation with custom streak calculation algorithms.

TetriC — Multiplayer Terminal Game | *C, POSIX Socket Programming, Multi-threading (pthreads), Non-blocking I/O*

[GitHub ↗](#)

- Engineered networked multiplayer console Tetris in raw C using POSIX TCP sockets, non-blocking I/O, and client-server architecture.
- Designed custom binary packet serialization protocol, real-time board state synchronization, and ANSI terminal rendering engine.

OPEN SOURCE & HONORS

romankh3/image-comparison | *Open Source Contributor (PR #255) — Maven Central Library (1,000+ Stars)*

2025

- Contributed enhancements to Maven Central visual regression testing library; improved diffing accuracy and expanded JUnit 5 test suites.

Competitive Programming Achievement | *Codeforces & LeetCode*

2022 – Present

- Solved **1,500+ algorithmic problems** across Codeforces, LeetCode, and regional programming contests; regular participant in National Collegiate Programming Contests (NCPG).

TECHNICAL SKILLS

Languages:	C++, C, Java, Python, TypeScript, JavaScript, SQL, HTML/CSS
Frontend & Mobile:	React, Next.js 16, React Native / Expo, Tailwind CSS, JavaFX
Backend & Cloud:	Cloudflare Workers & R2, Node.js, PostgreSQL (Neon), SQLite, MySQL, Prisma, Drizzle ORM, REST APIs
DevOps & Tools:	Git, GitHub, Linux / Bash, Socket Programming, Docker, Vercel, Maven, JUnit 5, VS Code